Worksheet - lesson 1: Actors and Actions (Subjects and Actions)

Principles:

1. Put actions in verbs
2. Put characters in subjects
3. Keep subjects near verbs

Exercises (revise these sentences for clarity using the principles of this lesson):

1. Tyrosine phosphorylation by activated JAKs of cytokine receptor cytoplasmic domains then provides binding sites for the Src-homology-2 domain of the STAT proteins

Ans: Activated JAKs then phosphorylate tyrosine residues in the cytoplasmic domains of cytokine-receptors, providing binding sites for the Src-homology-2 domains of STAT proteins.

2. We subjected yeast to 20 min of amino acid deprivation and made ribosome-footprint and mRNA-abundance measurements.

Ans: We deprived yeast of amino acids for 20 min, then measured ribosome footprints and mRNA abundance.

3. The assumptions that all sites evolve at one of two evolutionary rates (conserved and nonconserved) that these rates are uniform across the genome that sites evolve independently conditional on whether they are in conserved or nonconserved regions, and that the phylogenetic models for conserved and nonconserved regions have the same branch length proportions, base compositions, and substitution patterns.

all represent oversimplications of the complex process of sequence evolution in eukaryotic genomes.

Ans: These assumptions all oversimplify how sequences evolve in eukaryotic genomes: all sites evolve at one of two rates (conserved or nonconserved), these rates are uniform across the genome, sites evolve independently based on their region type, and phylogenetic models for both region types share the same branch-length proportions, base compositions, and substitution patterns.

4. The number of different mechanisms that may exist for cells to interpret morphogens, and the importance of design features such as feedback or local cell-cell communication is unclear.

Ans: It remains unclear how many mechanisms cells use to interpret morphogens or how important design features such as feedback and local cell-cell communication are.

5. Furthermore, the application of new technologies to further understand the biology of the adipocyte, including location analysis, global DNase hypersensitivity, high-throughput RNA-interference screens and computational strategies promises to enhance our knowledge of this once-neglected cell.

Ans: Furthermore, applying new technologies - including location analysis, global DNase hypersensitivity mapping, high-throughput RNA-interference screens, and computational strategies - promises to advance our understanding of this once-neglected cell type.

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6. Indeed in the mouse the sequences of large noncoding RNAs which probably have no 3' polyA tail, were reconstructed from the fragments of truncated cDNAs.
Ans: Indeed researchers reconstructed the sequences of large noncoding RNAs in the mouse from fragments of truncated cDNAs, although these RNAs probably lack a 3' polyA tail.

7. Localized fluctuations in substitution rate are widely employed to draw inference concerning the phenotypic significance of genomic sequence.
Ans: Researchers widely use localized fluctuations in substitution rate to infer the phenotypic significance of genomic sequences.

8. A number of promoters exhibited significant positive correlations between the footprinting estimated distribution of K and nucleosome score estimated from T-cells.
Ans: Several promoters showed significant positive correlations between the footprinting-estimated distribution of K and the nucleosome score estimated from T-cells.

9. In this study, we subjected seven different primate species to comparative analysis of the radial distribution pattern of human chromosome 18- and 19-homologous chromatin by three-dimensional fluorescence in situ hybridization.
Ans: In this study, we used three-dimensional fluorescence in situ hybridization to compare the radial distribution patterns of human chromosome 18- and 19-homologous chromatin across seven primate species.

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Worksheet - lesson 2: Cohesion, Coherence, and Emphasis

Principles:

1. Put new information last
2. Use passive voice judiciously
3. Make sure the first and last sentences of a paragraph match

Exercises (revise these sentences for clarity using the principles of this lesson):

1. These data were then subjected to hierarchical clustering using the Pearson correlation as the distance metric.

Ans: We then clustered these data hierarchically, using Pearson correlation as the distance metric.

2. We designated the putative GRB target genes as the human TFs that were under a HCNE density peak and whose orthologs were in conserved synteny with an array of HCNEs in human:zebrafish alignments. Density peaks and syntheric regions were downloaded from the Ancora Genome Browser.

Ans: We designated as putative GRB target genes those human TFs beneath an HCNE density peak whose orthologs showed conserved synteny with an array of HCNEs in human-zebrafish alignments. We downloaded density peaks and syntenic regions from the Ancora Genome Browser.

3. By analyzing all of the available mouse cDNA sequences, it was found that the mouse genome encodes at least 44050 distinct transcriptional units; a transcriptional unit comprises all the RNAs that share a common exonic sequence transcribed from the same genomic strand.

Ans: By analyzing all available mouse cDNA sequences, we found that the moese genome encodes at least 44000 distinct

transcriptional units - each comprising all RNAs that share a common exonic sequence transcribed from the same genomic strand.

4. Improvements are expected in the predictive power of all the scores being computed on multispecies alignments.
Ans: The predictive power of all scores computed from multispecies alignments is expected to improve.

5. With the development of molecular embryology and the coming of the post-genomic era, the molecular mechanisms of morphological evolution have recently begun to be elucidated.
Ans: The development of molecular embryology and the post-genomic era have recently enabled researchers to begin elucidating the molecular mechanisms of morphological evolution.

6. To assess the degree of enrichment of enhancer activities in predicted tissues, we compared the relative frequency of enhancers for each of the three tissues examined here with a background set of 528 previously tested sequences predicted to be developmental enhancers on the basis of extreme sequence constraint that were not associated with a prior tissue specificity prediction.
Ans: To assess enhancer enrichment in predicted tissues, we compared the relative frequency of enhancers in each of the three tissues examined here with that in a background set of 528 previously tested sequences that were predicted to be developmental enhancers based on extreme sequence constraint but had no prior tissue-specificity prediction.

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Worksheet - lesson 3: Concision and Simplicity

Principles:
1. Omit needless words
2. Prefer simple words
3. Use simple subjects
4. Use adjectives frugally

Exercises (revise these sentences for clarity using the principles of this lesson):

1. Most used CTSSs in the PMA time-course are observed to be, on average, approximately 450 bp away from an H3K9 AC island.
Ans: Most CTSSs in the PMA time-course are about 450 bp from an H3K9 AC island.

2. One of the complexes formed by the hematopoietic transcription factor Gata1 is a complex with the Ldb1 and Tal1 proteins.
Ans: The hematopoietic transcription factor Gata1 forms a complex with Ldb1 and Tal1.

3. The mechanistic connections among levels of phenotypic variation, for example, between spatial or temporal patterns of gene expression and morphology, determine how development constrains or channels evolution.
Ans: The connections between gene-expression patterns - spatial or temporal - and morphology determine how development constrains or channels evolution.

4. The scrutiny by natural selection is held to be greater for exons than their flanking intron sequences due to the explicit encoding of molecular function by exons.
Ans: Natural selection acts more strongly on exons than on flanking introns because exons encode molecular function.

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5. A phylogenetic tree based on the number of differentially expressed genes between species recapitulates their known phylogeny.
Ans: A phylogenetic tree based on differentially expressed genes recapitulates the known species phylogeny.

6. Modifications of the trichome pattern that have been examined in insects (resulting from laboratory-induced mutations or evolutionary diversification) are so far all attributable to changes in expression of svb.
Ans: All trichome pattern modifications examined in insects - whether from laboratory-induced mutations or evolutionary diversification - stem from changes in svb expression.

7. It is important to note for some of our later discussion that we permit a block to have only one row.
Ans: We allow each block to contain only one row, a constraint relevant to our later discussion.

8. These sequences are generally among the least constrained in the genome, so they constitute a plausible neutral standard accounting for regional variation in mutation and recombination rates.
Ans: Because these sequences are among the least constrained in the genome, they serve as a plausible neutral standard for regional variation in mutation and recombination rates.

9. The reasons for S_n differing among data sets are of considerable interest. Recent studies show that genes encoding proteins involved in developmental and transcriptional regulation tend to have highly constrained CRMs. In contrast, the extensive studies in the HBB gene complex...
Ans: The reasons for differences in S_n among data sets are important. Recent studies show that genes encoding proteins involved in developmental and transcriptional regulation tend to have highly constrained CRMs. In contrast, the extensive studies in the HBB gene complex...

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